Can eliminating dark energy resolve the Hubble tension? A uniform test of void and backreaction cosmologies against supernovae, baryon acoustic oscillations, and the CMB acoustic scale

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The timescape cosmology of Wiltshire dispenses with dark energy, attributing the apparent cosmic acceleration to the backreaction of inhomogeneities and the associated wall–void clock-rate difference, and has been claimed to ease the Hubble tension. We test this on present public data along three lines. First, treating the published timescape determinations of the dressed Hubble constant as fixed, the early- and late-Universe values are mutually consistent within the model; but because the global dressed value lies below *both* the Planck and the local distance-ladder numbers, this recasts the tension as a scale-dependent measurement artifact rather than a reconciliation near the local H_0. Second, reanalysing the Pantheon+ supernovae from first principles, we find the model-comparison verdict is strongly sensitive to the covariance treatment: spatially-flat ΛCDM is favoured under the full statistical+systematic covariance, and the timescape preference reported by proponents emerges only when the off-diagonal terms are discarded or the sample is restricted to low redshift. Third, calibrated against DESI baryon-acoustic-oscillation distances and the Planck acoustic scale, timescape fits the geometry worse than ΛCDM and prefers a present-day void fraction in significant tension with the supernova-preferred value. This shape tension recurs across the wider void and backreaction family, none of which reaches the local H_0, and it is not relieved by injecting the measured line-of-sight void structure. A free smooth expansion history does, however, reconcile the three datasets, locating the shape tension in the tracker's one-parameter rigidity rather than in any dark-energy-free description. We conclude that, on present public data, eliminating dark energy through inhomogeneity does not resolve the Hubble tension.

I. INTRODUCTION

The Hubble tension is the disagreement between late- and early-Universe determinations of H_0. The local Cepheid+type Ia supernova (SN Ia) measurement of SH0ES is $H_0 = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [1]; the Planck CMB value under spatially-flat ΛCDM is $H_0 = 67.36 \pm 0.54\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [2]; the discrepancy is quoted at 5σ. JWST rules out Cepheid crowding as the cause at 8.2σ [3]. The picture is, however, calibrator-dependent: the tip-of-the-red-giant-branch (TRGB) ladder of the Chicago–Carnegie Hubble Program gives a markedly lower $H_0 \simeq 68$–$70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [4, 24]—only ~ 1–1.5σ above Planck—and baryon-acoustic inverse-distance-ladder values remain near 68 [5]. The strong 5σ tension is therefore largely specific to the SH0ES Cepheid calibration; the target a late-Universe model must reach is ~ 69 if one adopts the TRGB ladder, or 73 for Cepheids.

A logically distinct class of resolution questions the homogeneous Friedmann–Lemaître–Robertson–Walker (FLRW) interpretation itself. The timescape cosmology of Wiltshire [7, 8] models the present, void-dominated Universe as a two-phase Buchert average [9]: spatially-flat "walls" (bound structures, where observers sit) and negatively-curved "voids." A quasilocal gravitational-energy and clock-rate gradient means a wall observer fitting FLRW infers "dressed" parameters distinct from the volume-average "bare" ones; apparent acceleration follows without dark energy. Recent Bayesian comparisons against Pantheon+ report strong evidence for timescape over ΛCDM [16, 17], reviving the question of whether the framework also addresses the Hubble tension—a possibility argued directly in a recent essay of Wiltshire [45], which frames the dynamical spatial curvature of the void-dominated Universe as a potential resolution. This paper reaches the opposite conclusion on present public data, so we engage that argument explicitly below.

The popular framing conflates two claims that are not equivalent: **(A)** within timescape the early- and late-Universe dressed Hubble constants *agree*; and **(B)** timescape *resolves* the 67-versus-73 tension. We show (Sec. III) that (A) holds at high precision while (B) holds only in a specific sense. We then ask whether the supernova evidence is robust (Sec. IV); we find the preference is contingent on the covariance treatment.

Originality. The framework relations of Sec. II and the published best-fit parameters are due to Wiltshire and collaborators and are reproduced for self-containment. New here are five contributions: (i) the explicit dressed-H_0 tension statistic and local-excess cross-check of Sec. III; (ii) the independent Pantheon+ reanalysis of Sec. IV, localising the timescape preference to the discarding of the off-diagonal covariance, together with a direct reproduction under the proponents' cosmology-independent covariance (Sec. IV D); (iii) the joint BAO+CMB calibration and the SN-versus-

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BAO+CMB void-fraction split (Sec. V); (iv) a uniform, same-harness test of the wider void and backreaction family (Sec. VI); and (v) a line-of-sight test injecting the measured 2M++ density field along real supernova sightlines (Sec. VIII). All code is available (Sec. XI). The scope limit is one of comparison depth, not model coverage: the family test is a geometric, ΔBIC-level distance comparison on a single CMB acoustic point and a standard sound horizon, not a per-model fully-marginalised Bayesian evidence or a full CMB-power-spectrum fit.

II. TIMESCAPE FRAMEWORK

We use only what is needed; see [7, 8] for derivations. The void volume fraction $f_v(t)$ grows to a present value f_{v0}. The lapse $\bar{\gamma} \equiv dt/d\tau_w$ relates volume-average time t to wall proper time τ_w, and in the late-time tracker solution the dressed and bare Hubble parameters are related by [8, eqs. (B7),(B9)]

$$H_0 = \frac{4f_{v0}^2 + f_{v0} + 4}{2(2 + f_{v0})}\,\bar{H}_0, \qquad \bar{\gamma}_0 = \tfrac{1}{2}(2 + f_{v0}). \qquad (1)$$

The bare deceleration parameter stays positive while the dressed one turns negative for $f_v \gtrsim 0.59$. Best fits give dressed $H_0 \approx 61.7$ and bare $\bar{H}_0 \approx 50\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [10, 11].

III. DRESSED-HUBBLE CONSISTENCY AND RESOLUTION-VS-RECASTING

We quantify agreement by $T_{AB} = |H_0^A - H_0^B|/\sqrt{\sigma_A^2 + \sigma_B^2}$.

ΛCDM reference. For SH0ES versus Planck, $T = 5.68/\sqrt{1.04^2 + 0.54^2} = 4.85$ ($\approx 5\sigma$ as quoted) [1].

Timescape. The CMB-calibrated dressed value is $61.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (± 0.79 stat, ± 4.88 sys) [12] and the SN-calibrated value is $61.7^{+1.2}_{-1.1}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [8, 10]; the residual is $0.7\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, so $T^{\mathrm{TS}} = 0.50$ (stat) or 0.14 (stat⊕sys). The same statistic that registers 4.85σ for the ΛCDM reading registers $\lesssim 0.5\sigma$ for the timescape reading, establishing claim (A).

Local-excess cross-check. Timescape predicts that a wall observer infers an apparent H_0 biased high below the $\sim 100\,h^{-1}$Mpc homogeneity scale, peaking near the $\sim 30\,h^{-1}$Mpc dominant-void scale [8, 13]. The excess required to reconcile SH0ES with the dressed asymptote, $\delta_{\mathrm{req}} = 73.04/61.7 - 1 = 0.184 \pm 0.028$, lies within the predicted ∼17–22% apparent-Hubble variance (local maxima $\sim 75\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) [8]. This is a magnitude-consistency argument: the SH0ES Hubble-flow sample extends to $z \sim 0.15$, beyond the homogeneity scale, so the account relies additionally on the cosmic-rest-frame choice, for which [14] report evidence favouring the Local Group frame.

Resolution versus recasting. The global dressed value ($\approx 61\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) lies below *both* 67.4 and 73.0. Timescape does not split the difference near 70; it reinterprets both canonical values as biased relative to a lower global average. This is a *recasting* of the tension as an FLRW-interpretation artifact, with consequences for consistency with inverse-distance-ladder BAO [5] not yet established for timescape at ΛCDM precision.

A fresh late-time anchor. The $\lesssim 0.5\sigma$ agreement above pairs the CMB-calibrated $61.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ with the 2008-era SN-calibrated $61.7\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, both tied to early-Universe or literature calibrations. Executing decisive test (ii) of Sec. X—replacing the literature late value with a fresh SH0ES-calibrator-anchored dressed H_0 measured inside the identical machinery (validation gate: the flat-ΛCDM anchored fit recovers $H_0 = 73.5 \pm 1.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_{m0} = 0.33$, matching [19])—yields a fresh timescape dressed $H_0 = 73.0 \pm 1.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($f_{v0} = 0.853$, $M_B = -19.24$). The validation gate is a calibration consistency check rather than independent confirmation of the specific local-variance mechanism: any smooth low-redshift distance relation reduces to $d_L \simeq cz/H_0$ for the Cepheid-host supernovae at $z \lesssim 0.01$ regardless of shape parameter, so recovering $H_0 = 73.5$ verifies the anchoring alone—the mechanism itself is what Sec. VIII tests, with a null result. This is expected, not a falsification: anchoring to the local Cepheid ladder pins H_0 to precisely the value a wall observer is predicted to infer biased high, and the required fractional excess $\delta_{\mathrm{req}} = H_0^{\mathrm{local}}/H_0^{\mathrm{dressed}} - 1 = 0.197 \pm 0.023$ lies inside the predicted 17–22% apparent-Hubble-variance window. The $\approx 9\sigma$ gap between this local-anchored 73.0 and the CMB-calibrated dressed asymptote 61.0 is thus consistent with that predicted variance window rather than an anomaly; but the paper's own directional line-of-sight test (Sec. VIII), injecting the measured 2M++ density field along real supernova sightlines, finds no such directional void signal, so this excess is best read as the same variance/covariance-driven low-redshift fitting behaviour diagnosed there, not a confirmed physical local-void expansion-rate bias. Claim (A) accordingly rests on this expansion-variance argument—a recasting—and never on an H_0 agreement at 61; making the late value a fresh measurement makes that dependence explicit.

IV. INDEPENDENT REANALYSIS OF PANTHEON+ SUPERNOVAE

A. Data, models, likelihood

We use the Pantheon+ release [18, 19]; the cosmology sample is the $N = 1580$ SNe with $z_{\mathrm{HD}} > 0.01$ that are not Cepheid calibrators ($0.010 < z < 2.26$; the exact selection is `IS_CALIBRATOR` = 0 and $z_{\mathrm{HD}} > 0.01$ with no further quality cuts, which accounts for the small difference from the ~ 1590 sometimes quoted). For flat ΛCDM, $d_L = (1 + z_{\mathrm{hel}})(c/H_0)\int_0^{z_{\mathrm{HD}}} dz'/E(z')$.

For timescape we implement the dressed luminosity distance from [15, App. A] verbatim: $d_L = (1+z)^2 d_A$, $d_A = c\,t^{2/3}[F(t_0) - F(t)]$ with

$$F(t) = 2t^{1/3} + \frac{b^{1/3}}{6} \ln \frac{(t^{1/3}+b^{1/3})^2}{t^{2/3} - b^{1/3}t^{1/3} + b^{2/3}} + \frac{b^{1/3}}{\sqrt{3}} \tan^{-1} \frac{2t^{1/3} - b^{1/3}}{\sqrt{3}\, b^{1/3}}, \qquad (2)$$

$1+z = 2^{4/3} t^{1/3}(t+b)/[f_{v0}^{1/3} \bar{H}_0\, t(2t+3b)^{4/3}]$, $b = 2(1-f_{v0})(2+f_{v0})/(9 f_{v0} \bar{H}_0)$, and (1). Only f_{v0} or Ω_{m0} shapes the Hubble diagram. We compare by ΔBIC (equal nuisance counts), adopting the Kass–Raftery scale [21], marginalising the absolute-magnitude/H_0 offset analytically.

B. Pipeline validation

Three independent checks: (i) our flat-ΛCDM fit gives $\Omega_{m0} = 0.333 \pm 0.018$, matching the published Pantheon+ value [19]; (ii) $F'(t)$ reproduces the integrand $2/[(2+f_v)t^{2/3}]$ to machine precision; (iii) our numerical low-z Taylor coefficient of $\mu_{\mathrm{TS}}(z)$ matches the analytic expression of [15] to $\sim 10^{-4}$ across $f_{v0} \in [0.6, 0.8]$.

C. Covariance sensitivity

We vary two axes. The apparent magnitude: the standard bias-corrected $m_{B,\mathrm{corr}}$ versus a raw SALT2 Tripp combination $m_B + \alpha x_1 - \beta c$ [20] with no FLRW-fiducial BBC bias correction. And the covariance: the full public statistical+systematic matrix C versus its diagonal. Results are in Table I and Fig. 1.

With the *full* covariance, ΛCDM is favoured for *every* magnitude definition (ΔBIC $= +3.9$ to $+5.1$; $+9.0$ if the bias correction is removed), and the timescape void fraction rails high ($f_{v0} \approx 0.85$–0.90, implying a strongly void-dominated dressed $\Omega_{m0} \approx 0.2$; the timescape posterior retains $\approx 21\%$ of its peak density at the $f_{v0} = 0.95$ prior edge, so this railed value is sensitive to the prior upper bound). A timescape preference (ΔBIC ≈ -8) appears *only* in the raw-Tripp+diagonal corner—i.e. only when the off-diagonal covariance is discarded. A fully marginalised statistical-only Tripp fit (fitting $\alpha, \beta, \sigma_{\mathrm{int}}$) gives ΔBIC $= -9.4$, consistent with this corner. The driver of any timescape preference is therefore the suppression of the off-diagonal covariance, not the magnitude standardisation alone. Figure 2 shows the near-degeneracy of the two distance–redshift relations (< 0.05 mag) that makes the comparison so sensitive.

Exact one-dimensional-quadrature Bayes factors remove any objection that the ΔBIC is a crude approximation. With the full statistical+systematic covariance, $\ln B(\Lambda\mathrm{CDM/TS}) = +1.3$ to $+1.8$ across a 2×2 prior grid (Kass–Raftery "positive", favouring ΛCDM; smaller than ΔBIC/2 because ΛCDM carries a $\sim$ 1-nat larger Occam factor[1]). By contrast our in-house statistical-only Tripp reduction, which drops the systematic off-diagonal covariance, gives $\ln B = 5.4$–5.9 *for* timescape ("very strong"). This matches the *magnitude* of the $\ln B > 5$ reported by [17], but by a distinct construction: their evidence is built on a *rebuilt* cosmology-independent covariance, under which our own whole-sample fit returns only $\ln B \approx -0.8$ at $z > 0.01$ (Sec. IV D)—mildly favouring timescape, not > 5. The lesson is the same (the sign and strength of the supernova evidence are set by the covariance treatment, not by the model), but the strong-for-timescape number comes from discarding the off-diagonal terms, which is not what the proponents do. A parametric bootstrap calibrates the frequentist significance: were timescape true, the observed supernova-only ΔBIC $= +5.1$ in ΛCDM's favour would arise with probability $p \approx 0.011$ (2.3σ; 4000 mocks drawn from the

TABLE I. Timescape vs flat ΛCDM on Pantheon+: ΔBIC (timescape$-\Lambda$CDM; < 0 favours timescape) across magnitude $\times$ covariance. ΛCDM wins under the full covariance regardless of magnitude; timescape wins only when the off-diagonal covariance is dropped.

Magnitude	Full cov	Diagonal
$m_{B,\mathrm{corr}}$ (standard, bias-corr.)	$+5.1$	$+4.9$
$m_{B,\mathrm{corr}}$, bias corr. removed	$+9.0$	$\cdots$
raw Tripp ($m_B + \alpha x_1 - \beta c$)	$+3.9$	-7.9

Best-fit $f_{v0} \approx 0.82$–0.90 throughout (vs $\Omega_{m0} = 0.333$ for ΛCDM); a fully marginalised stat-only Tripp fit gives ΔBIC $= -9.4$. The bias-correction-removed variant is evaluated under the full covariance only (its f_{v0} rails to the grid edge).

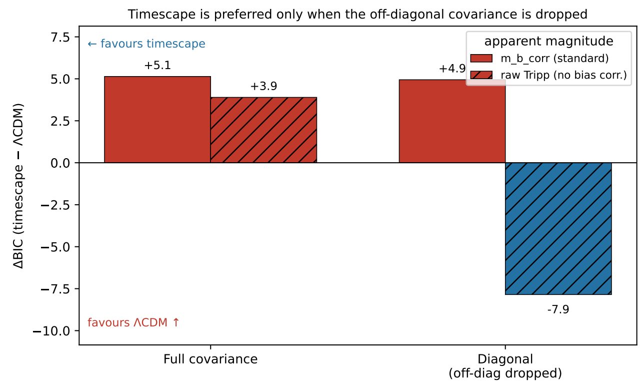

FIG. 1. Model-comparison verdict on Pantheon+ across magnitude (solid $= m_{B,\mathrm{corr}}$, hatched = raw Tripp) and covariance (full vs diagonal). Red favours ΛCDM, blue favours timescape. Only the raw-Tripp+diagonal case favours timescape; the full covariance favours ΛCDM for every magnitude definition.

[1] The maximised-likelihood ratio favours ΛCDM by $\Delta\chi^2/2 \approx 2.6$, but ΛCDM's prior-to-posterior compression costs ≈ 1 nat more than timescape's, so $\ln B \approx \Delta\chi^2/2 - 1 \approx 1.6$.

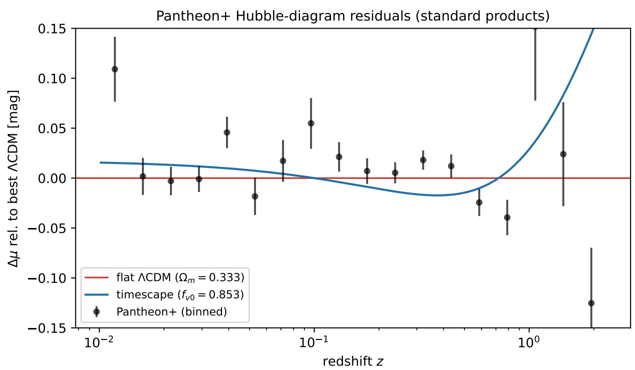

FIG. 2. Pantheon+ Hubble-diagram residuals relative to the best-fit flat ΛCDM (standard products). The best-fit timescape relation (blue) departs from ΛCDM by < 0.05 mag over the full range, explaining the covariance sensitivity.

TABLE II. Timescape versus flat ΛCDM under the cosmology-independent Seifert–Lane P+1690 covariance, as a function of the CMB-frame redshift cut $z_{\rm min}$ (frequentist profile likelihood; $\Delta\text{BIC} = \text{BIC}_{\rm TS} - \text{BIC}_{\Lambda\rm CDM}$ with $k = 9$ each, so $\Delta\text{BIC} = \Delta\chi^2$; both statistics < 0 favour timescape, > 0 favour ΛCDM). $\ln B$ profiles the nuisances and integrates the cosmological parameter (its 2×2-prior spread is ± 0.3). The maximum-likelihood f_{v0}, Ω_{m0} reproduce the published values of [16].

$z_{\rm min}$	N	f_{v0}	Ω_{m0}	ΔBIC	$\ln B$
0.010	1578	0.670	0.451	−0.5	−0.8
0.033	1169	0.675	0.433	+1.5	+0.3
0.055	1032	0.680	0.415	+4.9	+2.0

full stat+sys covariance, Gaussian tail approximation—a generalised-Pareto tail model gives 2.4σ).

D. Reproducing the cosmology-independent covariance

Our diagonal cases *discard* the off-diagonal covariance; they are not equivalent to the cosmology-independent covariance of [16, 17], which *rebuilds* the off-diagonal (FLRW-geometry- and peculiar-velocity-independent) terms. That covariance, with its P+1690 input vector, is now public,[2] so we run our timescape-versus-ΛCDM comparison directly under it, using the authors' Nielsen–Guffanti–Sarkar hierarchical likelihood (latent SALT2 population with profiled α, β and intrinsic dispersion, marginalised absolute magnitude) with our own distance code, which we verify is numerically identical to theirs; the likelihood itself we cross-check against an independent implementation to 10^{-13}.

Our frequentist maximum-likelihood fit reproduces the published values, $f_{v0} = 0.675$ and $\Omega_{m0} = 0.433$ [16] (their quoted maximum-likelihood void fraction, evaluated at $z_{\rm CMB} = 0.033$; the $z_{\rm min} = 0.033$ row of Table II), and recovers the reported dependence on the redshift cut (Table II): timescape is marginally favoured below the statistical-homogeneity scale ($\Delta\text{BIC} = -0.5$ at $z_{\rm min} = 0.01$, with $\ln B$ likewise mildly favouring timescape), there is no significant preference at $z_{\rm min} = 0.033$, and ΛCDM is favoured at $z_{\rm min} = 0.055$ ($\Delta\text{BIC} = +4.9$). This *reproduces*, in its maximum-likelihood void fraction and its redshift-cut trend, the finding of [16, 17]—a timescape preference concentrated below the homogeneity scale that weakens above it—rather than refuting it. Above the homogeneity scale the two analyses diverge in *sign*, however: our frequentist $\Delta\text{BIC} = +4.9$ at $z \geq 0.055$ favours ΛCDM, whereas [17] report a Bayesian $\ln B > 1$ still favouring timescape even at $z > 0.075$. What we reproduce is thus the maximum-likelihood parameters and the qualitative cut dependence, not the whole-sample Bayesian verdict, which above the homogeneity scale remains estimator- and prior-dependent.

Three caveats bound the interpretation. (i) Our frequentist $f_{v0} \approx 0.68$ at $z \geq 0.055$ lies 0.057 below the $f_{v0} = 0.737 \pm 0.029$ that [17] obtain by Bayesian nested sampling: this is $\approx 2\sigma$ measured against their posterior width alone, but only $\approx 1.3\sigma$ once our own profile width is combined in quadrature. Either way it is an estimator difference (frequentist maximum likelihood versus Bayesian posterior on a skewed profile), not a discrepancy—our value matches the frequentist maximum-likelihood estimate of [16]. (ii) Their headline $\ln B > 5$ is a fully-marginalised evidence for the whole sample ($z_{\rm min} \to 0$); our $\ln B$ profiles the nuisance parameters and integrates the single cosmological parameter by quadrature. Only the *nuisance* Occam factors cancel between models (identical nuisance structure); the single cosmological-parameter Occam factor does not—ΛCDM's Ω_{m0} prior compresses ≈ 1 nat more than timescape's f_{v0} prior—which is one reason we take ΔBIC as the primary statistic and $\ln B$ as a consistency check. (iii) The ΛCDM preference at $z \geq 0.055$ is mild-to-moderate ($|\Delta\text{BIC}| < 6$), not decisive. The picture is thus sharpened rather than overturned: under the cosmology-independent covariance the supernova preference is a low-redshift, scale-dependent effect, not a robust whole-sample preference for either model.

V. JOINT CALIBRATION: BAO AND THE CMB ACOUSTIC SCALE

To test whether the dressed $H_0 \approx 61\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ is consistent with the inverse distance ladder, we calibrate timescape against the DESI DR2 BAO distances [6] and the Planck acoustic scale $100\theta_* = 1.04109$ [2], using

[2] Covariance and input: Zenodo doi:10.5281/zenodo.12729746; pipeline and construction: `https://github.com/antosft/SNe-PantheonPlus-Analysis`.

the drag-epoch sound horizon r_d as the standard ruler. We compute the dressed transverse comoving distance $D_M(z)$, Hubble distance $D_H(z) = c/H(z)$, and $D_V(z)$ from the tracker solution and fit (f_{v0}, $\alpha \equiv c/(\bar{H}_0 r_d)$). Since r_d enters only the amplitude α, the redshift *shape*—and hence f_{v0}—is independent of the sound-horizon calibration.

The fit prefers $f_{v0} = 0.636^{+0.003}_{-0.004}$ and, with the standard $r_d = 147.1\,\mathrm{Mpc}$, a dressed $H_0 = 62.6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$: close to the dressed value of Sec. III, and recovering the independent CMB determination $f_{v0} = 0.627$ of [12], which validates the calculation. However the fit is poor, $\chi^2/\mathrm{dof} = 6.7$, versus 0.89 for flat ΛCDM ($\Omega_{m0} = 0.30$, $H_0 = 69\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ on the same data and pipeline); BAO alone prefer $f_{v0} = 0.68$. The BAO+CMB and supernova data prefer different void fractions (Fig. 3). We quantify the split three ways, which together bound its significance. (a) A profile-likelihood parameter-shift in f_{v0}—the rise in the jointly-minimised χ^2 above the sum of the separate minima, referred to χ^2 with one degree of freedom—gives, against the paper's own Pantheon+ fit, 6.6σ for the BAO+CMB pairing and 4.8σ for the CMB- and r_d-free BAO-only pairing (6.5σ and 4.4σ on DR1). This is an upper end: the supernova $\chi^2(f_{v0})$ profile is markedly non-parabolic, railing toward high f_{v0}, so its rise near the joint minimum is $\approx 1.9\times$ the Gaussian expectation and the $\sqrt{\Delta\chi^2}$ conversion overstates a nominal Gaussian significance. (b) A Gaussian parameter-difference built from the two profile widths at their minima, which does not inherit the non-parabolic tail, gives 4.7σ (BAO+CMB) and 3.6σ (BAO-only). (c) A distribution-free parametric bootstrap: of 4000 mocks drawn from a single-void-fraction timescape truth, none reproduces the observed SN-versus-BAO f_{v0} split (largest 0.12 versus the observed 0.21), a $\gtrsim 3.2\sigma$ floor by the rule of three.[3] The honest range is therefore ~ 3.2–7.5σ: a $\gtrsim 3.2\sigma$ distribution-free floor, 3.6–4.7σ on the Gaussian parameter difference, 4.8–6.6σ on the profile shift, and 7.5σ only under a tail model. A ladder over five supernova standardisations and four BAO treatments (from the DR1 ladder; DR2 only raises it) locates the softenings precisely. Including the $\sim 13\%$ CMB f_{v0} systematic [12] on the BAO+CMB side, a systematic that the r_d-independent BAO-only fit does not carry, drops the split to ~ 1.8–2.3σ whichever of the paper's own SN reductions is used, and to $\sim 1.2\sigma$ if the external cosmology-independent anchor $f_{v0} = 0.737$ is also substituted (this last adopts an assumed ± 0.02 uncertainty on the anchored SN void fraction; the published ± 0.029 is larger and lowers the figure slightly further, to ~ 1.1–1.2σ). Across the CMB-systematic-free, non-anchor subgrid (the paper's own SN reductions against the BAO-only fit) the split retains a 3.1σ floor. The

[3] A Gaussian or generalised-Pareto extrapolation of the mock tail gives $\approx 7.5\sigma$, but that is a tail-model figure beyond the mock support; we quote the distribution-free floor.

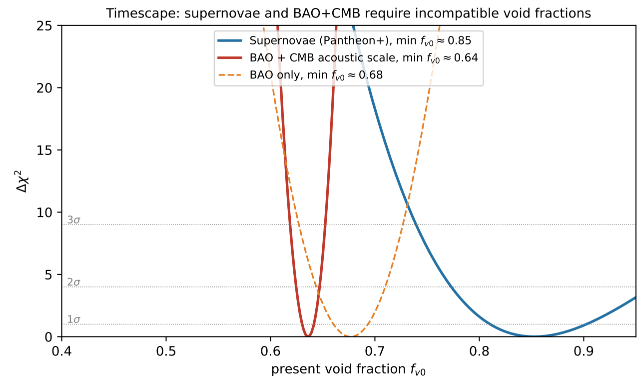

FIG. 3. $\Delta\chi^2$ for the present void fraction f_{v0} from supernovae (blue), BAO+CMB (red), and BAO alone (orange). The supernova and BAO+CMB minima are separated by $\Delta f_{v0} \approx 0.2$ with non-overlapping confidence regions: no single timescape model fits both.

BAO+CMB-alone model comparison gives ΔBIC(TS − ΛCDM) = +69.3 (+23.3 on DR1). We therefore report a robust tension between the SN- and BAO+CMB-preferred void fractions—largely r_d-independent (it concerns the distance–redshift shape), to be settled definitively only with a self-consistent timescape sound horizon; the timescape literature reports concordance across these probes [10, 12]. The CMB here enters only through the acoustic-scale point $D_M(z_*)/r_d$, not the full power spectrum; a self-consistent timescape r_d would shift H_0 but not the shape-driven f_{v0} tension, which the BAO-only fit confirms. These figures use DESI DR2; DR1 is the more conservative choice on every axis and we retain it as a sensitivity check.

A. Apparent acceleration and a comparison with evolving dark energy

Apparent acceleration is present in every best fit, so the comparison is not an artifact of one model lacking it: the dressed timescape deceleration parameter is $q_0 = -0.02$ to -0.04 (a weak apparent effect from the lapse, with the bare $\bar{q}_0 > 0$), versus $q_0 = -0.54$ for ΛCDM (joint fit). As a motivated alternative that instead *modifies* the acceleration history, we also fit the CPL evolving-dark-energy model $w(a) = w_0 + w_a(1-a)$ jointly to SN+BAO+CMB. It returns $w_0 = -0.87$, $w_a = -0.47$ (the thawing direction hinted at by DESI [5]), with $q_0 = -0.40$, but improves the fit by only $\Delta\chi^2 = 3.9$ for two extra parameters (ΔBIC = +10.8 versus ΛCDM), so it is not significantly preferred. Table III collects the joint fits: ΛCDM is the most economical, evolving dark energy a mild non-significant improvement in the DESI direction, and timescape strongly disfavoured (ΔBIC = +67) because no single void fraction fits SN and BAO+CMB.

TABLE III. Joint SN+BAO+CMB fits ($N = 1593$), identical harness. ΔBIC relative to ΛCDM; $q_0 < 0$ denotes acceleration. BAO data here are DESI DR1; the DR2 recalibration of Sec. V only sharpens the timescape disfavour.

Model	parameters	H_0	q_0	ΔBIC
ΛCDM	$\Omega_{m0} = 0.305$	68.6	-0.54	0
w_0w_aCDM	$\Omega_{m0} = 0.31, w_0 = -0.87, w_a = -0.47$	68.0	-0.40	$+10.8$
timescape	$f_{v0} = 0.64$	63.1	-0.02	$+67.1$

B. Simplifications

The calibration makes explicit approximations, none of which affect the central SN–BAO+CMB incompatibility (which is geometric and, in the BAO-only form, independent of r_d and the CMB): (i) the CMB enters only through the acoustic-scale point $D_M(z_*)/r_d$, not the full power spectrum or the (R, ℓ_A, ω_b) compression, and its error is floored at 0.05 ($\approx 2\times$ looser than Planck's $\sim 0.03\%$ determination of $100\theta_*$), a deliberate conservative inflation—tightening it only strengthens the timescape disfavour; (ii) we adopt the standard early-physics $r_d = 147.1$ Mpc rather than a self-consistent timescape sound horizon, which would rescale H_0 but not the shape-driven f_{v0}; (iii) the timescape tracker is the late-time attractor, so $D_M(z_*)$ evaluates it beyond the regime where it is an accurate description, though the comoving distance to last scattering is dominated by the low-redshift ($z \lesssim$ few) portion of the integral where the tracker holds; (iv) radiation is neglected in the late-time distance integrals; (v) ΛCDM is taken spatially flat.

VI. THE WIDER VOID AND BACKREACTION FAMILY

To test whether the timescape result is special, we applied the identical SN+BAO+CMB harness to the principal dark-energy-free inhomogeneous cosmologies (Table IV). Each distance–redshift relation was implemented from primary sources and validated against published or analytic limits; the SN absolute-magnitude offset and the BAO ruler $c/(H_0 r_d)$ were marginalised as before.

The outcome is uniform. (i) The LTB giant-void model (constrained GBH [25]) reproduces SN acceleration with a deep central void but, like timescape, prefers a *deeper* void for SNe than for BAO+CMB (joint ΔBIC $\approx +717$); its central H_0 reaches only ~ 55–59, and it is independently excluded by the kinetic Sunyaev–Zel'dovich effect [27, 38]. (ii) The Räsänen peak/backreaction model [28, 29] is parameter-free and, by its authors' own evaluation, does not even produce acceleration ($\bar{q} > 0$, $|\Omega_{\mathcal{Q}}| < 0.04$); promoting the backreaction to a free parameter, SNe demand values beyond its physical reach and the joint fit gives ΔBIC $\approx +72$ (above timescape's $+67$ once its extra backreaction parameter is charged the Occam penalty the same harness applies to w_0w_aCDM and LTB). (iii) The Buchert backreaction "morphon" template [30] fits SNe but fails BAO+CMB catastrophically (ΔBIC $\sim +1.6 \times 10^3$). (iv) The Szekeres model [31] has no closed-form isotropic distance and could not be reliably fit here; the literature verdict (kSZ-capped void depth, residual $\sim 3\sigma$ tension [27]) is that it does not resolve the tension; a recent effective Λ-Szekeres model fittable to the local Universe with Cosmicflows-4 finds that accounting for local structure *increases* the H_0 tension, shifting the best-fit H_0 by $\approx 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [60]. (v) The local KBC void [32, 33]—which *retains* dark energy—leaves BAO+CMB untouched and earns no statistical preference (a fixed literature-depth void worsens the joint fit by ΔBIC $\approx +165$, growing monotonically with depth, so the no-void ΛCDM limit is preferred); a void deep enough to reach 73 is both deeper than observed and disfavoured by the supernova Hubble-flow shape [27, 34]. The wider literature is actively divided on the local void as an H_0 mechanism: ΛLTB fits to SN+BAO+CMB find an underdensity does not solve the tension [53], the required void is emptier than ΛCDM structure allows [54], and direct distance tracers prefer a void far smaller (< 70 Mpc) than the KBC fiducial [55]; against this, local-void models have been argued to reduce a twenty-year BAO-compilation tension from 3.3σ to 1.1–1.4σ [56], to survive DESI DR2 without definitive exclusion [57], and to track the observed decline of the inferred H_0 with redshift [58]. Our harness verdict here—a fixed literature-depth KBC void worsens the joint fit—stands as computed, but the row's underlying literature status is contested on present data.

Every dark-energy-free void/backreaction model thus shows the same SN-versus-BAO+CMB parameter split as timescape, and none reaches the local H_0: the failure is generic to the class, not specific to one model. The ΔBIC magnitudes from this geometric harness are approximate (single CMB acoustic point, standard r_d), but the qualitative verdict is robust and, for the LTB/Szekeres subclass, reinforced by the independent kSZ exclusion. One recent family member we did not test is the ηCDM "emergent dark energy from structure formation" of [59], an ensemble-averaged closure claiming to ease both the H_0 and $f\sigma_8$ tensions; implementing its distance–redshift relation in this harness is left to future work.

VII. PRIOR DARK-ENERGY-FREE RESOLUTIONS AND THEIR STATUS

The proposal that apparent acceleration—and, by extension, the Hubble tension—reflects the inhomogeneity of the late Universe rather than a new energy component has a long lineage, and our same-footing test sits within a literature of single-model studies. We summarise that landscape here, separating the proponents' claims from the principal critiques, so that our uniform survey (Sec. VI) and the void-family Table IV can be

TABLE IV. Uniform SN+BAO+CMB test of void/backreaction cosmologies on the same harness. "DE-free" marks models that eliminate dark energy. ΔBIC is relative to ΛCDM (joint $\chi^2 = 1402$); larger is more disfavoured. None resolves the tension.

Model	DE-free	joint ΔBIC	verdict
ΛCDM (ref.)	no	0	—
w_0w_aCDM	no	+11	no (mild DESI-direction hint)
Timescape	yes	∼+67	no
LTB void (GBH)	yes	∼+717	no (+kSZ-excluded)
Räsänen peak	yes	∼+72	no (predicts no acceleration)
Buchert morphon	yes	∼+1600	no
Szekeres	yes	intractable	no (literature + kSZ)
KBC local void	no	0 (no-void lim.)	no (needs unobserved deep void)

read in context.

The timescape program. Timescape grew out of Wiltshire's resolution of the cosmological averaging problem [7, 8], built on the Buchert spatial-averaging formalism [9], with the bare/dressed distinction as its conceptual core: a wall observer fitting FLRW infers parameters that differ from the volume average. Leith, Ng & Wiltshire [10] reported the first multi-probe concordance fit (SNe+CMB+BBN mutually consistent without dark energy at a low dressed H_0), Duley, Nazer & Wiltshire [11] added a radiation fluid to reach recombination, and Nazer & Wiltshire [12] fitted the CMB acoustic peaks, quoting a dressed $H_0 \simeq 61\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ with large (∼ 8–13%) systematics. On supernovae the verdict has been sensitive throughout: Smale & Wiltshire [13] found timescape preferred for MLCS2k2-reduced samples but disfavoured for SALT-reduced ones, identifying the light-curve fitter as the dominant systematic, and Dam, Heinesen & Wiltshire [15] found timescape statistically indistinguishable from flat ΛCDM on JLA with marginal apparent acceleration. The review of Wiltshire [35] frames dark energy as a coarse-graining artifact and rests part of the case on a sub-$100\,h^{-1}$Mpc Hubble-flow variance and a claimed Local-Group-frame uniformity [14]—a cosmic-rest-frame inference that remains contested. An independent N-body group-catalogue test of the predicted Hubble-flow variation [64] found the data to prefer ΛCDM over a differential-expansion approximation to timescape, which Wiltshire [65] rebutted on the grounds that the Newtonian rescaling omits the model's key geometrical features. More recently, an essay of Wiltshire [45] has argued that the dynamical spatial curvature of the void-dominated Universe can potentially resolve the Hubble tension—a qualitative mechanism claim that the present quantitative multi-probe test does not bear out. The recent revival sharpens the supernova claim: Lane et al. [16] and Seifert et al. [17] reanalyse Pantheon+ with a covariance constructed to be independent of FLRW geometry and peculiar-velocity corrections and report very strong Bayesian evidence ($\ln B > 5$) for timescape over ΛCDM, with the preference persisting beyond the statistical-homogeneity scale. We test these claims directly (Sec. IV D): running our comparison under their now-public cosmology-independent covariance, we reproduce their published maximum-likelihood void fraction ($f_{v0} = 0.675$) and their reported redshift-cut dependence—timescape marginally favoured below the statistical-homogeneity scale, our frequentist statistic favouring ΛCDM above it where their Bayesian evidence still mildly favours timescape ($\ln B > 1$ at $z > 0.075$). The supernova preference is thus real but scale-dependent, a low-redshift effect rather than a robust whole-sample result, and its Bayesian magnitude ($\ln B > 5$) reflects their fully-marginalised evidence for the whole sample where ours is a nuisance-profiled approximation.

Backreaction and giant voids. A parallel tradition seeks acceleration directly from spatial averaging: Räsänen's mechanism [36], in which faster-expanding voids coming to dominate the volume drive an effective acceleration, which his own quantitative peak-model evaluation [28] then found too weak, and the Buchert "morphon" repackaging of backreaction as an effective scalar [30]. A scaling-solution closure with dynamical spatial curvature has already been fit to type Ia supernovae [52] and found statistically indistinguishable from ΛCDM on the supernova data by an Akaike criterion, its best-fit parameters checked for consistency against independently published CMB and BAO constraints. Whether backreaction can be dynamically significant is the subject of a long, still-open exchange: Green & Wald argue the metric is everywhere close to FLRW so backreaction is trace-free and cannot source acceleration [22, 47], while an eleven-author rebuttal [23], and a subsequent analysis of the formalism's assumptions [48], dispute that it establishes backreaction to be negligible and question the restrictiveness of its assumptions. The dispute is partly one of gauge: spatial averaging yields a gauge-dependent, hence ambiguous, backreaction magnitude [49]. Recent first-principles GR computations on the real local Universe find the kinematical backreaction small ($\lesssim 1\%$) even as the integrated spatial curvature reaches ∼ 10% [42], numerical-relativity averaging on Newtonian-gauge-like hypersurfaces likewise finds it negligible while the curvature varies at the ∼ 10% level [50], and full numerical-relativity void statistics supply a 10–30% void-overexpansion that motivates the void-fraction parameter [41]; a first observational assessment over an extended redshift range cannot yet tightly constrain the total backreaction either way [51]. The LTB/Szekeres giant-void line [25] fits SNe with a deep central void but is bounded sharply from outside: radial BAO and a low-H_0/fine-tuning dilemma [26], a multi-probe exclusion [37], and—decisively—the kinetic Sunyaev–Zel'dovich effect, an unavoidable consequence of large radial inhomogeneity that rules out adiabatic Gpc voids [38]. The distinct *local*-void (H_0-outflow) idea, including the deep KBC void [32, 33], is separately disfavoured: the local Hubble-flow shape shows no outflow of the required size [34] and the supposed void resolution is an artifact of the redshift range used to calibrate H_0 [27]. This local-void question is under active debate on present data (Sec. VI). Distinct from all of these averaging and void routes are proposals that the apparent acceleration is itself an observational artifact—of a large-scale bulk-flow anisotropy in Pantheon+ [61], or of a progenitor-age luminosity drift that, once corrected, points to deceleration [62]—which place the effect in the supernova data rather than in cosmic inhomogeneity.

The recent (2023–26) context. Several developments have renewed interest. The DESI inverse-distance-ladder gives $H_0 \simeq 68\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and hints at evolving dark energy ($w_0 > -1$, $w_a < 0$) [5], which the no-dark-energy programs read as room for an inhomogeneity-driven effective equation of state. The tension's apparent magnitude has also softened on the calibrator axis: the TRGB ladder of the Chicago–Carnegie Hubble Program gives $H_0 \simeq 68$–70, only ~ 1–1.5σ from Planck [4, 24], so the 5σ figure is largely SH0ES-Cepheid-specific. Against this, the review/ranking literature has consistently placed the inhomogeneity class below early-time new physics: the field's solution catalogues assign it a weaker verdict [39], and metric-based rankings find late-Universe and local mechanisms cannot resolve H_0 under joint BAO+SN constraints [40]. The community-wide CosmoVerse white paper—co-authored by timescape proponents—reviews the tension landscape, the potential systematics, and the candidate new physics, and sets out the coming decade's objectives for addressing the discordances [46].

This work in context. The studies above are overwhelmingly single-model: each proponent paper fits one framework, and each critique bounds one mechanism. Our contribution is a *uniform* test—the identical SN+BAO+CMB harness, with the same likelihood, marginalisation, and model-comparison statistic applied across the whole dark-energy-free family (Sec. VI)—which makes the recurring SN-versus-BAO+CMB parameter split visible as a property of the class rather than of any one model. It complements, rather than supersedes, the detailed single-model analyses: where the proponents retain a cosmology-independent supernova covariance whose recent public release now makes reproduction possible, and where the critiques deploy probes (kSZ, radial BAO) outside our distance-only harness, our test adds the same-footing comparison that the existing literature lacks.

VIII. DIRECTIONAL AVERAGING OR MODEL RIGIDITY? A LINE-OF-SIGHT TEST

Timescape applies one averaged distance–redshift curve to every sightline, yet the local void content is *measured* at low redshift. We tested directly whether injecting it changes the supernova verdict, using the reconstructed 2M++ galaxy density field [43] along the line of sight to each of the 573 Pantheon+ supernovae (460 distinct) inside the reconstruction's reliable radius ($\lesssim 200\,h^{-1}$Mpc, $z \lesssim 0.067$); redshifts are placed with $z_{\rm CMB}$, not the peculiar-velocity-corrected $z_{\rm HD}$, to avoid regressing the data on the same 2M++ field.

Three findings, each a null for the "directional-averaging" hypothesis. First, a generalised-least-squares regression of the Hubble residuals on the per-sightline void fraction, under the full covariance, is consistent with zero once calibrated against a sky-rotation null (-0.9σ); the naive regression error understates the scatter by $\approx 2.7\times$ because it ignores the covariate's spatial coherence and the covariance's coherent modes, so an apparent $\sim 2\sigma$ signal is sub-threshold and does not survive. Second, a one-parameter per-sightline extension of the tracker (an expansion-variance coupling to the measured void excess) improves the fit by only $\Delta\chi^2 \approx 0.6$ and leaves $\hat{f_{v0}}$ at 0.85: random rotations of the field help the fit as much on average (rotation-null $p \approx 0.6$). Third, the maximal per-sightline magnitude modulation allowed by the data is $\lesssim 0.05$ mag, a few percent of the per-supernova error and steeply weighted to $z \lesssim 0.1$, because $\mathrm{Var}(F) \sim L_{\rm void}/D(z)$ shrinks along longer beams. Measured structure therefore adds no information at Pantheon+ precision: the single-parameter *average* is not the bottleneck, and the low-redshift origin of the supernova preference is a variance/covariance effect rather than a cosmological signal.

Two further checks locate the tension. Externally, the supernova-preferred $f_{v0} \approx 0.85$ exceeds the observed present-day void volume fraction under any structure-based definition (numerical-relativity watershed void statistics bracket ≈ 0.50–0.62 [41]), whereas $f_{v0} \approx 0.64$ sits just above the top edge of that bracket and coincides with timescape's own Planck fits [12]. And a free, spline-parametrised smooth expansion history fit jointly to supernovae, BAO, and the CMB acoustic scale matches ΛCDM ($\Delta\chi^2 = 10$ for five extra parameters, $p = 0.065$; the Wilks p is approximate here, as ΛCDM is not nested in the spline family, but conservatively so—it does not inflate the spline's improvement over ΛCDM) and sits far below the timescape tracker ($\Delta\chi^2 \approx 67$). A smooth history *can* reconcile the three datasets; the timescape tracker cannot. The SN-versus-BAO+CMB split is thus the rigidity of the one-parameter averaged history (one f_{v0} forced to serve a low-redshift supernova shape and the BAO+CMB geometry at once), not a failure to resolve directional structure, and not an irreducible supernova–BAO shape disagreement. This is a robustness result: it strengthens, rather than overturns, the recasting interpretation of the Discussion below.

A companion analysis [44] pursues the same diagnosis in the full dressed-geometry solver rather than the kinematic spline used above: freeing the void-fraction history $f_v(z)$ from the one-parameter tracker recovers an ΛCDM-quality joint SN+BAO+CMB fit, isolating the tracker's rigidity, not the dressed-geometry framework itself, as the cause of the combined-fit failure. That reconciliation is a kinematic reading (the forced history need not satisfy the Buchert integrability conditions) and the analysis is ongoing; code at https://github.com/szhygulin/free-history-timescape.

IX. DISCUSSION

Three conclusions follow. First, interpreted within timescape the early- and late-Universe *dressed* H_0 agree at ≈ 61, and the local excess is of the predicted magnitude (a fresh SH0ES-calibrator-anchored late-time fit returns 73.0 ± 1.0, consistent with the predicted variance window rather than 61, though the directional line-of-sight test of Sec. VIII finds no confirming void signal); but this is a recasting, not a reconciliation near 70, and the global value lies below Planck. Second, the supernova model-comparison verdict is contingent on the covariance treatment: it favours ΛCDM under the full public covariance and timescape only when the off-diagonal terms are removed. Third, when timescape is required to fit supernovae, BAO, and the CMB acoustic scale *together*, the SN- and BAO+CMB-preferred void fractions differ ($\Delta f_{v0} \approx 0.1$–$0.2$) and timescape fits the BAO+CMB geometry worse than ΛCDM; the split is significant across estimators (~ 3.2–7.5σ; profile shift, Gaussian parameter difference, and distribution-free bootstrap floor), softening to ~ 1.2–2.3σ once the CMB f_{v0} systematic and the external cosmology-independent SN anchor are included (see Sec. V). The early/late dressed-H_0 consistency of Sec. III is thus genuine, but whether one parameter set also describes the combined distance data is unsettled on present evidence, pending the self-consistent sound horizon and cosmology-independent covariance. Notably the same SN-versus-BAO+CMB split recurs across the whole void/backreaction family (Sec. VI), suggesting it is generic to dark-energy-free inhomogeneous models rather than specific to timescape; a recent geometric analysis reaches a parallel conclusion for late-time $w(z)$ dark energy, finding that supernovae and BAO constrain Ω_{m0} through orthogonal channels that disagree [63], so the tension is not idiosyncratic to the void closure.

These results sit within known open problems: the timescape CMB fit [12] matches the acoustic-peak structure by adapting an ΛCDM template rather than from first principles; BAO and growth constraints are far less developed than for ΛCDM; and whether backreaction can be dynamically significant remains disputed, with the estimated magnitude sensitive to gauge and averaging choice [22, 23, 49, 50].

X. WHAT A DECISIVE TEST REQUIRES

(i) The cosmology-independent, FLRW-geometry- and peculiar-velocity-independent Pantheon+ covariance of Seifert et al. is now public and we have run it under our harness (Sec. IV D): the supernova preference is scale-dependent—marginally timescape below the statistical-homogeneity scale (ΔBIC $= -0.5$), ΛCDM above (ΔBIC $= +4.9$ at $z \geq 0.055$)—reproducing the proponents' redshift-cut trend and their maximum-likelihood void fraction. A fully-marginalised Bayesian evidence, versus our nuisance-profiled approximation, would sharpen the low-z magnitude but not the sign; that is what remains. (ii) A joint re-fit of Pantheon+ and Planck within timescape with the dressed H_0 *free*, and a self-consistent timescape sound-horizon/BAO calibration, so early and late H_0 are genuine independent measurements; the calibrator-anchored late-time half of this test is carried out in Sec. III, leaving the self-consistent sound horizon. (iii) A direct measurement of the radial profile of the apparent H_0 versus scale, to test the predicted expansion variance that the local-excess cross-check assumes.

XI. CONCLUSIONS

Within timescape the early- and late-Universe dressed Hubble constants agree to $\lesssim 0.5\sigma$ (versus $\approx 4.9\sigma$ in ΛCDM), and the local excess is of the predicted void-scale-variance magnitude (though the directional line-of-sight test of Sec. VIII finds no confirming void signal)—but the model recasts rather than reconciles the tension, collapsing both canonical values to $\approx 61\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, below Planck. Our independent Pantheon+ reanalysis finds ΛCDM favoured under our own full-covariance reductions (ΔBIC $= +3.9$ to $+5.1$; exact $\ln B = +1.3$ to $+1.8$), with a timescape preference appearing when the off-diagonal covariance is discarded (stat-only $\ln B = 5.4$–5.9 for timescape). Run under the proponents' now-public cosmology-independent covariance, our fit reproduces their published maximum-likelihood values and their scale-dependent result: timescape marginally favoured below the statistical-homogeneity scale (ΔBIC $= -0.5$ at $z > 0.01$), ΛCDM above it (ΔBIC $= +4.9$ at $z \geq 0.055$). The supernova preference is thus real but a low-redshift, scale-dependent effect, not a robust whole-sample preference. A joint calibration against DESI DR2 BAO and the Planck acoustic scale fits worse than ΛCDM ($\chi^2/\mathrm{dof} = 6.7$ vs 0.89) and prefers a void fraction ($f_{v0} \simeq 0.64$) in tension with the supernova value (0.74–0.85) at ~ 3.2–7.5σ across estimators (see Sec. V), softening to ~ 1.2–2.3σ once the CMB f_{v0} systematic and the external cosmology-independent SN anchor are included. The dressed-H_0 numbers *are* reconciled within timescape ($\approx 61\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$); what remains open is whether one parameter set also fits the combined SN+BAO+CMB distances. This last incompatibility is a rigidity of the one-parameter tracker rather than an irreducible shape mismatch: a free spline-parametrised smooth expansion history reconciles the three datasets at ΛCDM quality ($\Delta\chi^2 \approx 10$, $p = 0.065$, versus $\Delta\chi^2 \approx 67$ for the tracker; Sec. VIII), leaving open whether a non-tracker dark-energy-free history could do better—a question the companion analysis [44] pursues.

A uniform test of the wider void/backreaction family (Sec. VI) finds the same SN-vs-BAO+CMB split in every dark-energy-free model and none reaching the local H_0, with the LTB subclass independently excluded by the kSZ effect. Finally, the tension's magnitude is calibrator-dependent: the TRGB ladder gives $H_0 \simeq 68$–70, ~ 1–1.5σ from Planck, so the 5σ tension is largely an artifact of the SH0ES Cepheid calibration. On present public data, eliminating dark energy via inhomogeneity does not resolve the Hubble tension.

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DATA AND CODE AVAILABILITY

This work uses the public Pantheon+ release [18, 19] (`https://github.com/PantheonPlusSH0ES/DataRelease`). The analysis code (timescape and ΛCDM distance modules, the magnitude×covariance decomposition, the validation tests, and the figure scripts) is public at `https://github.com/szhygulin/timescape-hubble-tension`. The DR2 BAO+CMB headline (χ^2/dof $=$ 6.7 versus 0.89 for ΛCDM; the ~ 3.2–7.5σ SN-versus-BAO+CMB split, across estimators) is backed by `probes_out/dr2.json`; `results_baocmb_dr1.json` is the DR1 cross-validation/sensitivity check.

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